

A Square and a Cube — Full Test

Mixed questions from all four concepts. Section A is interactive; Sections B and C are written. Time: 45 minutes ·
Total: 28 marks. Answers on the separate key.

Objective (1 mark each)

Q1 · MCQ**[1]**

Which of the following is a perfect square?

(a) 48 (b) 64 (c) 72 (d) 80

Q2 · MCQ**[1]**

$\sqrt{196} = ?$

(a) 13 (b) 14 (c) 15 (d) 16

Q3 · MCQ**[1]**

The units digit of the cube of 17 is:

(a) 1 (b) 3 (c) 7 (d) 9

Q4 · MCQ**[1]**

$\sqrt[3]{343} = ?$

(a) 6 (b) 7 (c) 8 (d) 9

Q5 · FILL IN THE BLANK**[1]**

The number of natural numbers between 7^2 and 8^2 is:

Q6 · TRUE / FALSE**[1]**

1000 is a perfect square.

True False

Q7 · MCQ

[1]

Which of the following is **NOT** a perfect cube?

- (a) 27 (b) 64 (c) 100 (d) 125

Q8 · ASSERTION-REASON

[1]

Assertion (A): 441 is a perfect square.

Reason (R): $441 = 21^2$.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q9

[2]

Find $\sqrt{2025}$ by prime factorisation.

Q10

[2]

Find the units digit of the cube of (i) 29 (ii) 31.

Q11

[2]

Is 2800 a perfect square? Justify using prime factorisation.

Q12

[2]

Express 64 as a sum of consecutive odd numbers.

Long answer (3 marks each)

Q13

[3]

By what smallest number should 392 be multiplied to make it a perfect square? Find the square root of the result.

Q14

[3]

By what smallest number should 1080 be multiplied to make it a perfect cube? Show your working.

Q15

[3]

Find the smallest square number that is divisible by each of 4, 9 and 10.

Q16

[3]

2401 students are to be seated so that the number of rows equals the number of students in each row. How many rows are there?

Total: 28 marks · Check your work against the **answer key**.